Identifying Congenital Heart Disease in the Newborn Nursery

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KEYWORDS

• Newborn • Congenital heart disease • Cyanosis • Pulse oximetry screening

KEY POINTS

- Critical congenital heart disease can lead to severe morbidity or death if not detected early.
- Prenatal considerations and clinical assessment based on symptoms and signs of cardiac disease are the first step in determining the need for ancillary tests.
- Universal pulse oximetry screening has helped improve detection rates of critical congenital heart defects.
- Ancillary tests performed based on clinical assessment aid timely diagnosis of critical congenital heart defects.
- Future considerations include the use of point of care ultrasound, electronic stethoscopes, and telemedicine services.

INTRODUCTION

Congenital heart defects (CHDs) are the most common type of birth defects, affecting approximately 1% of live births in the United States.[1,2] These defects range from acyanotic, hemodynamically insignificant lesions, such as small atrial and ventricular septal defects and bicuspid aortic valves, to critical CHD (CCHD). CCHD comprises about 25% of CHD cases and are life-threatening lesions that require catheter-based intervention or cardiac surgery during the neonatal period. Advancements in fetal echocardiography and universal pulse oximetry screening have significantly improved the early recognition and delivery planning for CCHD. However, a small percentage of infants remain undiagnosed. Early diagnosis can be further enhanced by incorporating

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Abbreviations	
CCHD	critical CHD
CHDs	congenital heart defects
POCUS	point of care ultrasound

historical aspects and physical signs alongside pulse oximetry screening, allowing for the identification of CCHD before signs of decompensation occur.

Prenatal Considerations

Fetal echocardiogram may be beneficial for pregnancies with an increased risk of CHD. Fetal echocardiogram is routinely recommended for certain maternal risk factors, such as diabetes, exposure to specific medications or substances during pregnancy, and abnormal findings during routine prenatal ultrasounds.[3,4] A history of previous pregnancies with CHDs or a familial history of CHD confers a 2% to 3% increased risk of having infants with CHD.[5,6] Family history of heritable conditions with CHD manifestations may also prompt fetal echocardiogram, include bicuspid aortic valve with left-sided lesions, 22q11 deletion, Marfan syndrome, long QT syndrome, and anti Ro and anti La antibodies (SSA/SSB)-positive maternal connective tissue disease (placing infant at risk of neonatal congenital heart block). Advanced maternal age can also predispose infants to certain genetic conditions, such as chromosomal trisomies (trisomy 13, 18, and 21) and sex chromosome aberrations (such as monosomy X), which are often associated with CHDs. Positive prenatal screening for any of these conditions can prompt fetal echocardiogram assessment.

Fetal echocardiogram by a pediatric cardiologist can offer early diagnosis and facilitate prenatal counseling and delivery preparations. However, it may detect only 80% to 90% of CHD, and it is not cost-effective as a universal screening tool. Accessibility of this subspecialty care may also be challenging for some patients.[7]

SYMPTOMS RELATED TO CARDIAC DISEASE IN THE NEWBORN

Common clinical presentations of neonatal cardiac disease include poor feeding, cyanosis, cool extremities, or tachypnea (Fig. 1).[8] A detailed history can help pediatric clinicians determine the etiology of the presentation.

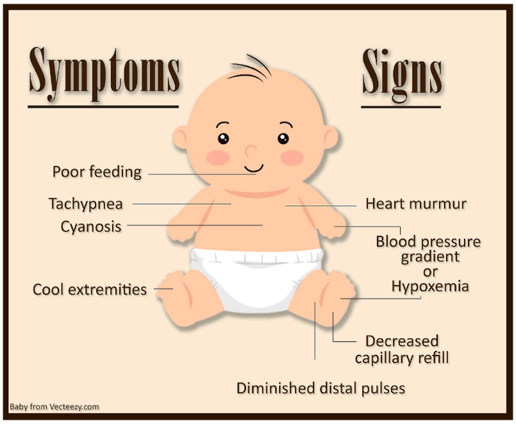

Fig. 1. Common presenting symptoms and signs of congenital heart disease in neonates. (Figure courtesy- Dr Carter Oster.)

Poor Feeding

Feeding requires increased cardiac output in an infant, and thus, it may trigger or exacerbate symptoms of cardiac disease. Infants with myocarditis, cardiomyopathy, or obstructive heart disease can present with poor feeding as an initial manifestation due to poor cardiac output. Infants with a large ventricular septal defect can present with signs of exertion and exhaustion associated with feeds (tachypnea, sweating, and prolonged feeding time) at around 4 to 6 weeks of age due to the large shunt causing pulmonary overcirculation at the expense of adequate cardiac output. Heart conditions that lead to decreased gut perfusion can lead to feeding intolerance. In coarctation of aorta after ductal closure, the infant may present with poor feeding due to decreased flow beyond the level of the coarctation.

Cyanosis

Central cyanosis is a bluish discoloration of mucus membranes and skin which is noted in the lips, tongue, cheeks, gums, and chest, and needs to be evaluated. This needs to be differentiated from peripheral cyanosis or acrocyanosis of the newborn where cyanosis can be present in the perioral venous plexus, hands, and feet. During a general examination, central cyanosis can indicate CHD or pulmonary disease. Features that help differentiate between pulmonary and cardiac causes include the response to oxygen (pulmonary causes typically improve), work of breathing (typically increased in pulmonary causes of cyanosis), and the presence of other cardiac signs such as impaired perfusion, murmurs, or abnormal heart sounds.

The causes of cyanosis in an infant include CHD, persistent pulmonary hypertension of the newborn, acquired or congenital primary lung disease, and infection including sepsis, hypothermia, and methemoglobinemia. In infants with primary CHD, typically a history of respiratory distress is not elicited. Infants with ductal-dependent CHD may have a benign birth history and present with cyanosis a few hours to days after birth, associated with increased physical effort (feeding and crying). This new onset cyanosis occurs after ductal closure due to decreased pulmonary blood flow. Cyanosis from birth may be noted with critical right heart obstructive lesions (pulmonary atresia, tricuspid atresia, Tetralogy of Fallot, and critical pulmonary stenosis), left heart obstructive lesions (critical aortic stenosis, interrupted aortic arch, and hypoplastic left heart syndrome), or mixing lesions (transposition of great arteries, truncus arteriosus, and total anomalous pulmonary venous return).

Cool Extremities

Patients presenting with decreased systemic perfusion can present with cool extremities as an initial symptom. Delivery history of intra-amniotic infection, elevated temperature of the birth parent during labor, and prematurity are risk factors for neonatal sepsis. Placental abruption can cause critical neonatal anemia. It is important to obtain family history of endocrine or metabolic abnormalities. With obstructive heart disease and lesions with ductal-dependent systemic blood flow, patients are usually stable in the first few hours of life and later develop signs of decreased systemic perfusion such as cool extremities, acidosis, and hemodynamic shock.

Tachypnea

Tachypnea due to cardiac disease typically occurs due to excessive pulmonary blood flow from a significant left to right shunt and typically manifests over a few days or weeks of life with a drop in pulmonary vascular resistance. Infants with this type of tachypnea are often referred to as having "comfortable tachypnea," that is, tachypnea without significant increased work of breathing.

SIGNS RELATED TO CARDIAC DISEASE IN THE NEWBORN

A systematic physical examination can reveal subtle signs of congenital heart disease. For patients with possible CHD, heart, skin, pulmonary, and abdominal examinations are especially important.

Hypoxemia

Hypoxemia is defined as insufficient oxygen at the tissue level, which is usually measured by pulse oximetry and arterial blood gas analysis.[9] This can occur due to various causes such as hypoventilation, ventilation-perfusion mismatch, cyanotic heart disease, or impaired oxygen-delivering capacity (ie, severe anemia, methemoglobinemia, and so forth). In an infant presenting with hypoxemia, response to supplemental oxygen may help differentiate between pulmonary and cardiac etiologies. If hypoxemia persists despite correction of pulmonary causes, a cardiac evaluation is warranted.

Heart Murmur

A normal cardiac examination includes splitting of the second heart sound. A single S2 can be suggestive of pulmonary or aortic atresia. A widely split single S2 associated with a systolic ejection murmur can be suggestive of an atrial septal defect. A slight parasternal lift along the left sternal border may be seen due to normal right ventricular dominance. A holosystolic murmur at the left sternal border radiating all over the precordium usually indicates a ventricular septal defect. Similarly, a continuous murmur at the left upper sternal border could indicate the presence of a patent ductus arteriosus along with bounding pulses can indicate hemodynamic significance. Typically, a soft systolic ejection murmur loudest at the left sternal border radiating to axilla is associated with peripheral physiologic pulmonary artery stenosis, which is a common innocent murmur in infants.

Blood Pressure Gradient

A discrepancy in upper and lower extremity blood pressure with systolic blood pressure gradient of greater than 20 mm Hg between upper and lower extremity is concerning for coarctation of the aorta or interrupted aortic arch.

Diminished Distal Pulses

An important part of the cardiac examination is palpation of the distal pulses. Diminished lower extremity pulses can be suggestive a coarctation of the aorta or interrupted aortic arch.

Decreased Capillary Refill

In addition, skin findings such as decreased capillary refill and mottled skin can be suggestive of decreased systemic perfusion and low cardiac output.

PULSE OXIMETRY AS A SCREENING TOOL

Some children with CCHD will not have any concerning signs or symptoms initially in the newborn nursery; however, they may have decreased oxygen saturations that are not severe enough to be clinically apparent. Pulse oximetry screening is a simple noninvasive tool, which has helped bridge some of the diagnostic gaps of undetected

CHD and has been proven to decrease infant mortality from CHD.[10] After excluding prenatally detected infants, it is estimated that nearly 50% to 70% of infants born with undiagnosed CCHD can be detected by pulse oximetry screening.[11,12] Pulse oximetry CCHD screening is most sensitive for hypoplastic left heart syndrome, pulmonary atresia, Tetralogy of Fallot, total anomalous pulmonary venous return, transposition of the great arteries, tricuspid atresia, and truncus arteriosus.[13]

Pulse oximetry screening is a noninvasive tool. The pulse oximetry reading in the right hand (preductal) and lower extremity (postductal) are obtained when the infant is greater than 24 hours old (or directly before pending discharge if <24 hours of age) and breathing room air. A passing result is a preductal and postductal measurement of greater than or equal to 95% with a difference of 3% or less between measurements. A saturation level less than 90% in either the preductal or postductal measurement should prompt immediate clinical assessment for both cardiac and noncardiac causes of hypoxemia. Oxygen saturations of 90% to 94% in either the preductal or postductal measurement, or a difference of greater than 3% between the prenatal and postnatal measurements, should prompt repeat screen in 1 hour. Persistent abnormal test results warrant postnatal echocardiogram.

In the United States, pulse oximetry screening for CCHD is universally recommended for all newborns after 24 hours of life. Abnormal CCHD results are reported to each state as part of the newborn screening program. State-specific workflows have differed, however, on specific criteria for repeating the test or obtaining an echocardiogram until the American Academy of Pediatrics released a unified workflow recommendation in 2024.[14] States also differ on whether CCHD diagnoses are reported to statewide birth defect registries. Overall, implementation of CCHD screening with universal pulse oximetry has been a cost-effective resource to detect CHD but studies have suggested that approximately 15% of infants may still be discharged from the hospital with undetected CCHD.[15]

A normal CCHD screening test at 24 hours does not imply that the infant's cardiac anatomy is typical. Functional closure of the ductus arteriosus occurs in the first hours to days in most newborns. Ductal-dependent lesions may not have precipitated hypoxemia at 24 hours of life if the ductus arteriosus has not yet closed.

CCHD screening is also not intended to detect acyanotic CHD. Ventricular septal defect is the most common CHD diagnosis in newborns, but may not cause sufficient mixing of oxygenated and deoxygenated blood to cause abnormally low pulse oximetry reading at 24 hours of life. Symptoms from pulmonary overcirculation or heart failure often do not appear for weeks in patients with this lesion.

Most interrupted aortic arch or coartation of the aorta is juxtaductal and may not become symptomatic until after the ductus arteriosus closes. Interrupted aortic arch/coartation of the aorta also encompasses a spectrum of left outflow tract obstruction severity, and milder cases may not cause sufficient change in postductal perfusion to be detected on clinical examination or pulse oximetry screening in the newborn nursery. Heart failure from left outflow tract obstruction may develop over the ensuing days to weeks, and ultimately precipitate symptoms that prompt diagnosis.

ANCILLARY TESTS THAT AID IN THE DIAGNOSIS WHEN CONGENITAL HEART DEFECT IS SUSPECTED

- *Arterial blood gas analysis:* It can help detect metabolic acidosis in a patient with significant respiratory distress, which can often be seen in patients with decreased systemic perfusion.

- *Hyperoxia test:* This is a very useful test to differentiate between cardiac and noncardiac causes of cyanosis. Initially arterial blood gas analysis is performed in room air. Following this, infants are provided 100% supplemental oxygen and arterial blood gas analysis is repeated in 10 minutes. If the PaO_2 is less than 100, it can be indicative of CCHD, and thus, further workup may be indicated.[15]
- *Electrocardiogram:* It can be useful in detecting arrhythmias but can be normal in many cardiac lesions at birth. There are certain findings that can be indicative of cardiac pathology such as atrial or ventricular hypertrophy, dextrocardia, abnormal axis deviation, or rhythm abnormalities.
- *Chest radiography:* Heart size, pulmonary blood flow, and lung parenchyma can be characterized. Severe right-sided cardiomegaly can be seen in Ebstein's anomaly (**Fig. 2**).[16] Loss of normal convexity in the left upper mediastinum due to the malposed pulmonary artery in D-transposition of great arteries (d-TGA) and is also called as an egg on a string appearance (**Fig. 3**).[17] Similarly in Tetralogy of Fallot, a small pulmonary artery and right ventricular hypertrophy can give a boot-shaped appearance of the heart. A snowman appearance on the chest radiograph has been described in diagnosis of total anomalous pulmonary venous return due to prominent superior vena cava on the right side of the trachea (**Fig. 4**).[18]
- *Transthoracic echocardiogram:* This is a definitive noninvasive test, which is the mainstay in the diagnosis of CHD.[19] Echocardiogram characterizes both structure and hemodynamics with the use of color and spectral Doppler. Availability can be limited in rural and remote areas.

FUTURE CONSIDERATIONS

Point of Care Ultrasound

Point of care ultrasound (POCUS) is increasingly being used in the neonatal intensive care unit for diagnosis and serial monitoring of patent ductus arteriosus, persistent

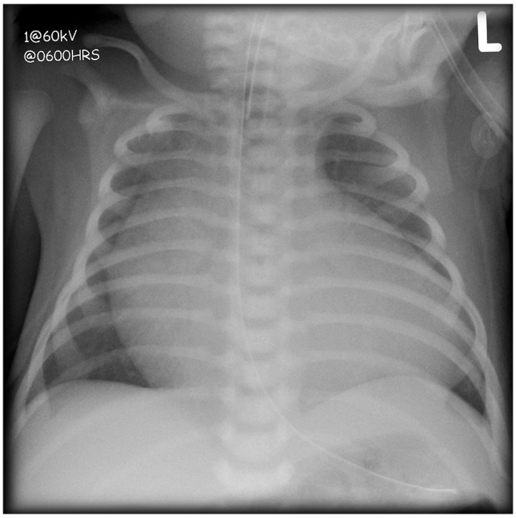

Fig. 2. Ebstein anomaly with severe right atrial enlargement. (Case courtesy of Frank Gaillard, Radiopaedia.org, rID: 7442.)

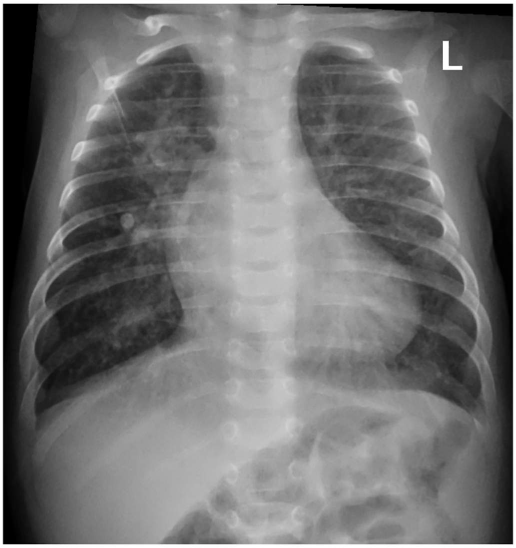

Fig. 3. Transposition of the great arteries with "egg on a string" appearance. (Case courtesy of Hani M. Al Salam, Radiopaedia.org, rID: 10685.)

pulmonary hypertension, identification and drainage of pericardial effusion, necrotizing enterocolitis, and intraventricular hemorrhage.[20] Opportunities for focused training to detect CCHD in the nursery, especially when access to cardiac care may be limited, may be a future area of development.

Electronic Stethoscopes and Artificial Intelligence Platforms

Electronic stethescopes are increasingly being used along with telemedicine to differentiate between innocent and pathologic murmurs.[21] Electronic stethoscopes such as Ausculband and Ekoduo generate a phonocardiogram in an audio format to be accessed by clinicians with the link.[22] Artificial intelligence algorithms have been

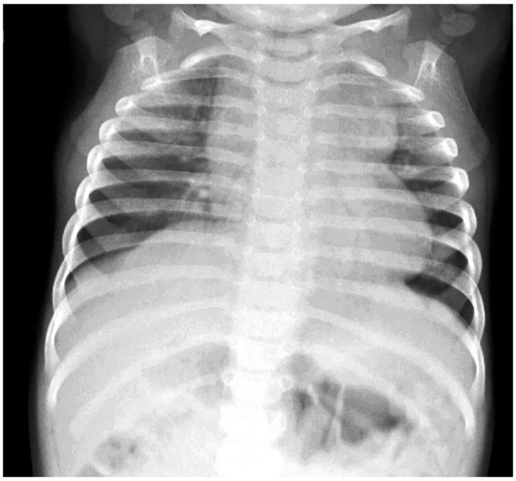

Fig. 4. Total anomalous pulmonary venous return with a classic "snowman" sign. (Case courtesy of Aditya Shetty, Radiopaedia.org, rID: 27800.)

created to analyze electronic stethoscope data to enhance clinician detection of cardiac murmurs in adult patients.[23] Further development of these technologies within pediatrics may be helpful for clinicians caring for newborns with limited in person access to pediatric cardiology services.

Telemedicine

With the introduction of electronic stethoscopes in the market and growing technological innovations, it has become easier for pediatric clinicians to utilize telemedicine technology to consult cardiologists. Various tertiary centers offer this option to clinicians and have trained sonographers available at local hospitals to perform an echocardiogram and provide interpretation remotely. This facilitates more effective triage. CCHD is prioritized for immediate care and transfer, while those with non-CCHD can avoid being transferred to the tertiary center and follow up on an outpatient basis.

SUMMARY

Early identification of CHD, especially CCHD, can be lifesaving. Fetal echocardiogram may facilitate prenatal diagnosis, counseling, and delivery preparation. Universal pulse oximetry screening at 24 hours of life aids the detection of CCHD, but it does not detect all CHD. Ultimately, it is a clinician's careful attention to presenting symptoms and physical examination signs, which facilitates timely diagnosis and treatment. Leveraging new technologies to enhance clinician detection of CHD may improve equitable access and allocation of resources. By improving timeliness and accuracy of the diagnosis, we hopefully will continue to improve clinical outcomes for children with CHD.

CLINICS CARE POINTS

- Prenatal screening has improved early recognition and delivery planning for CCHD.
- Pulse oximetry can detect 50% to 70% of infants born with undiagnosed critical congenital heart defects.
- Passing screening with pulse oximetry does *not* rule out possible CCHD. Approximately 15% of infants may still be discharged from the hospital with undetected CCHD despite prenatal and pulse oximetry screening.
- Assessment by pediatric clinicians based on symptoms and signs of cardiac disease and use of ancillary tests can help detect CHD in infants.
- Use of POCUS, electronic stethoscopes, and telemedicine services also aim to improve early detection of these defects.

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DISCLOSURE

The authors have no disclosures or declarations of interest.

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